

Abanoub Abdelmalak | ECE PhD Researcher | Hardware-Aware Machine Learning | ex-Siemens EDA

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Skills

AI & Machine Learning

- Deep Learning & Generative AI
- Natural Language Processing (NLP)
- Reinforcement Learning
- Bayesian Optimization
- PyTorch, TensorFlow, Scikit-learn
- LangChain & MCP Frameworks
- Computer Vision & Image Processing

Semiconductor & Quantum

- Quantum Error Correction (QEC)
- Qiskit, Stim, Amazon Braket
- Computational Lithography
- Optical Proximity Correction (OPC)
- Resolution Enhancement Techniques (RET)
- Metrology, CD-SEM

Software & Tools

- Python, C/C++, Java
- MATLAB, Perl, Tcl/Tk
- Linux Shell Scripting
- Git Version Control
- Cloud Computing (Azure, AWS)
- Web Development (Django, REST)
- High-Performance Computing (LSF)

Publications

An AST-guided LLM Approach for SVRF Code Synthesis

Abanoub E. Abdelmalak, Mohamed A. Elsayed, David Abercrombie, and Ilhami Torunoglu
First author; Published in the 1st IEEE International Conference on LLM-Aided Design (ICLAD 2025).

Siemens DI SW EDA

June 2025

Work Experience

Siemens Digital Industries Software (EDA)

Principal Product Architect

Cairo, Egypt | Shanghai, China

Oct 2022 – 2025

- Architected an **SVRF Copilot**, a domain specific coding language) to automate design rule decks development. Pipeline involved LLM, LangChain, and an MCP protocol with tool-aware reasoning. Published as first-author *IEEE ICLAD 2025*; reduced rule development time by **~40%**
- Designed an AI and image processing tool that utilizes Scanning Electron Microscope (SEM) images to automate process modeling; reduced manual review by **30%**.
- Led global delivery across China, Europe, and the US. Orchestrated an Agile infrastructure transition for major Chinese fab, transforming an initial one-month engagement into a 6-years partnership.
- Drove RET/OPC automation (pattern generation, metrology recipe creation) integrated with Calibre/LSF; improved run-time/throughput by **70%**.

Siemens Digital Industries Software (EDA)

IC Design Consultant

Shanghai, China

Aug 2017 – Oct 2022

- Developed an OPC model-building automation system covering test-pattern creation, data filtering/prep, SEM metrology recipe development, and OPC verification; accelerated calibration cycles and reduced manual steps by **90%**.
- Interfaced with physical metrology tools to auto-generate measurement; defined spatial pattern “uniqueness” via the mathematical entropy of extracted features.
- Architected a web-based recipe/version management platform with role-based access, improving security, tractability, and collaboration across teams.
- Implemented RET automation (pattern-coverage checks, DRM management) using C/C++ LAPI and ML-based feature extraction; increased coverage by 25% and cut setup time by 50%.
- Built computational models and test masks for technology ramps, meeting strict physical accuracy targets for hardware nodes from 90nm KrF to 5nm ArF.

Education

Georgia Institute of Technology

PhD Candidate in Electrical & Computer Engineering, Full Scholarship and Fellowship Recipient

Atlanta, GA, United States

Aug 2025 – Present

- Advised by Prof. Ali Adibi.
- Developing *Syndrome Learning*, an AI-driven framework for hardware-aware QEC targeting Penning trap ion crystals.
- Training explicit surrogate noise models on simulated and empirical hardware data to replace idealized i.i.d. noise assumptions with physical reality.

Georgia Institute of Technology

MSc. in Computer Science, Machine Learning Specialization, GPA: 3.8

Atlanta, GA, United States

Aug 2021 – Aug 2024

- Specialized in advanced machine learning algorithms through graduate research and coursework in Deep Learning, NLP, information security, and algorithmic design.
- Headed the Shanghai student chapter, supporting new students in the program and arranging regular meetings accordingly.

Ain Shams University

BSc. of Communication Systems Engineering, Optics Specialization, GPA: 3.85, BAC+5

Cairo, Egypt

Sept 2012 – July 2017

- Ranked first in the Optics track and seventh overall.
- Graduation Project: Low power phase locked loops for internet of things applications.

Projects

Domain Specific Language Copilot

Cairo, Egypt

Siemens DI SW EDA

2024 – 2025

- Initiated and delivered the minimal viable product that automatically generates Standard Verification Rule Format (SVRF) language code from natural-language descriptions.
- Integrated a chat interface using LangChain and connected the model to a language server for contextual suggestions. Integrated the system with Model Context Protocol (MCP) for agentic custom function execution.

Agent Solutions for EDA & Pattern-less PDK Verification

Cairo, Egypt

Siemens DI SW EDA

2024 – 2025

- Architected and implemented an end-to-end AI agentic flow to automate EDA verification tasks, orchestrating design checks, simulation, and optimization via LLMs and reinforcement learning.
- Developed a pattern-less framework for verifying Process Design Kit (PDK) rules and multiple MCP functions for PDK verification;
- Integrated these agents into existing workflows, enabling continuous validation of PDKs and accelerating verification cycles.

SEM Image Deep Learning Cleaning

Remote

Siemens DI SW EDA, Joint Customer Development

2024

- Designed and trained a deep neural network (DNN) system that automatically classifies physical SEM images by quality and structural feature type.
- Created a cross-platform QT interface to visualize and review large datasets; and automated generation of modeling files from cleaned data, reducing manual effort while improving fidelity.

Big Data Quantum Mechanics

Remote

Georgia Tech, Vertically Integrated Project; Supervised by Prof. Andrew J Medford

2023

- Conducted quantum mechanical simulations using **Quantum Espresso**, evaluating material adsorption energies and the impact of physical parameters on computational stability.
- Benchmarked the **Equiformer_v2** graph neural network against baseline models to evaluate its predictive accuracy for complex physical and molecular properties.

Reinforcement Learning for Stock Trading

Remote

Georgia Tech, Research Project

2022

- Architected a reinforcement learning system focusing on dynamic environment adaptation; designed custom reward functions and non-stationary exploration policies that successfully optimized sequential decision-making over baseline strategies.

Honors & Awards

Feb 2024: **Consultant Excellence**, Siemens DI SW EDA

Jan 2022: **Consultant Excellence**, Siemens DI SW EDA

Oct 2020: **Consultant Excellence**, Siemens DI SW EDA

Sep 2015: **Student Exchange Program**, DAAD Organization

Languages

Arabic: Native | **English:** Full Professional Proficiency | **French:** Elementary Proficiency | **German:** B1 Proficiency

Courses

Aug 2025: *Quantum Information and Computation*, IBM Quantum Curriculum

Nov 2023: **Introduction to Psychology**, Yale University, Coursera

Nov 2023: **Project Initiation: Starting a Successful Project**, Google, Coursera

Oct 2023: **Foundations of Project Management**, Google, Coursera

Sept 2023: **Fundamentals of Quantum Mechanics**, University of Colorado Boulder, Coursera

Sept 2023: **Agile Project Leadership**, LinkedIn Learning

Jun 2018: **Machine Learning**, Stanford University, Coursera